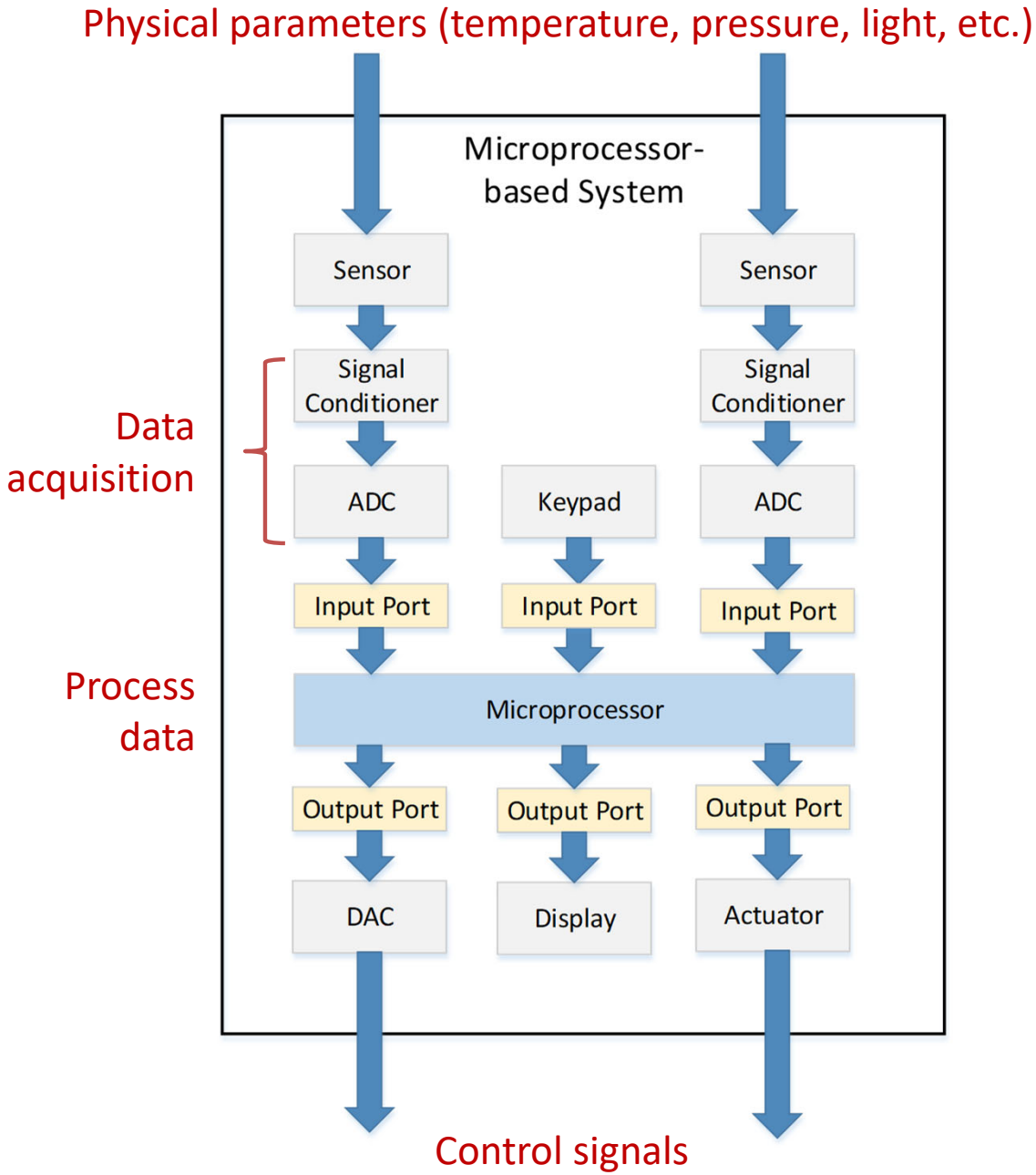
Recap

- Typical functions of an embedded system for automated control:
  - Monitor one or few physical parameters (temperature, pressure, light) and data acquisition
  - Process the acquired data
  - Control the physical parameters - system needs to generate signals



NANYANG TECHNOLOGICAL UNIVERSITY

SC2107/CE2107
Microprocessor System Design and Development

Sensors, Actuators and Signal Conditioning

Topics Covered

- The Measurement and Control Loop
- Sensors and Actuators
  - Temperature, Position, Motor
- Signal Conditioning Circuits
  - Amplifiers and Filters

Reference: Coughlin and Driscoll, "Operational Amplifiers and Linear Integrated Circuits", 6th Edition, Prentice Hall

The Measurement and Control Loop

Introduction

- Microprocessors are now used extensively in systems and applications that interact with surroundings through its I/O interfaces
- Real world signals are analogue in nature and they can be measured from sensors



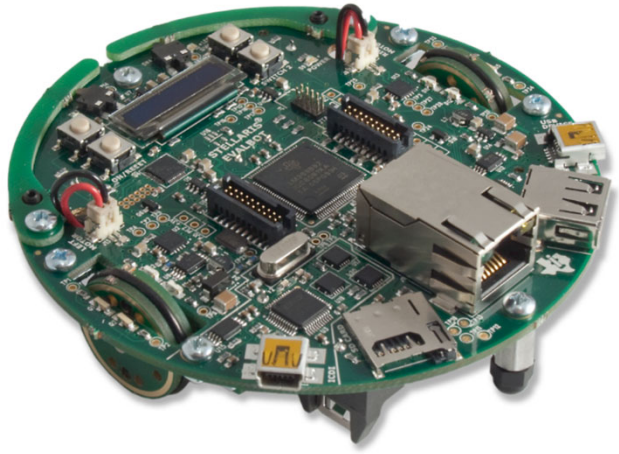
Smartphones







Play stations and Wii

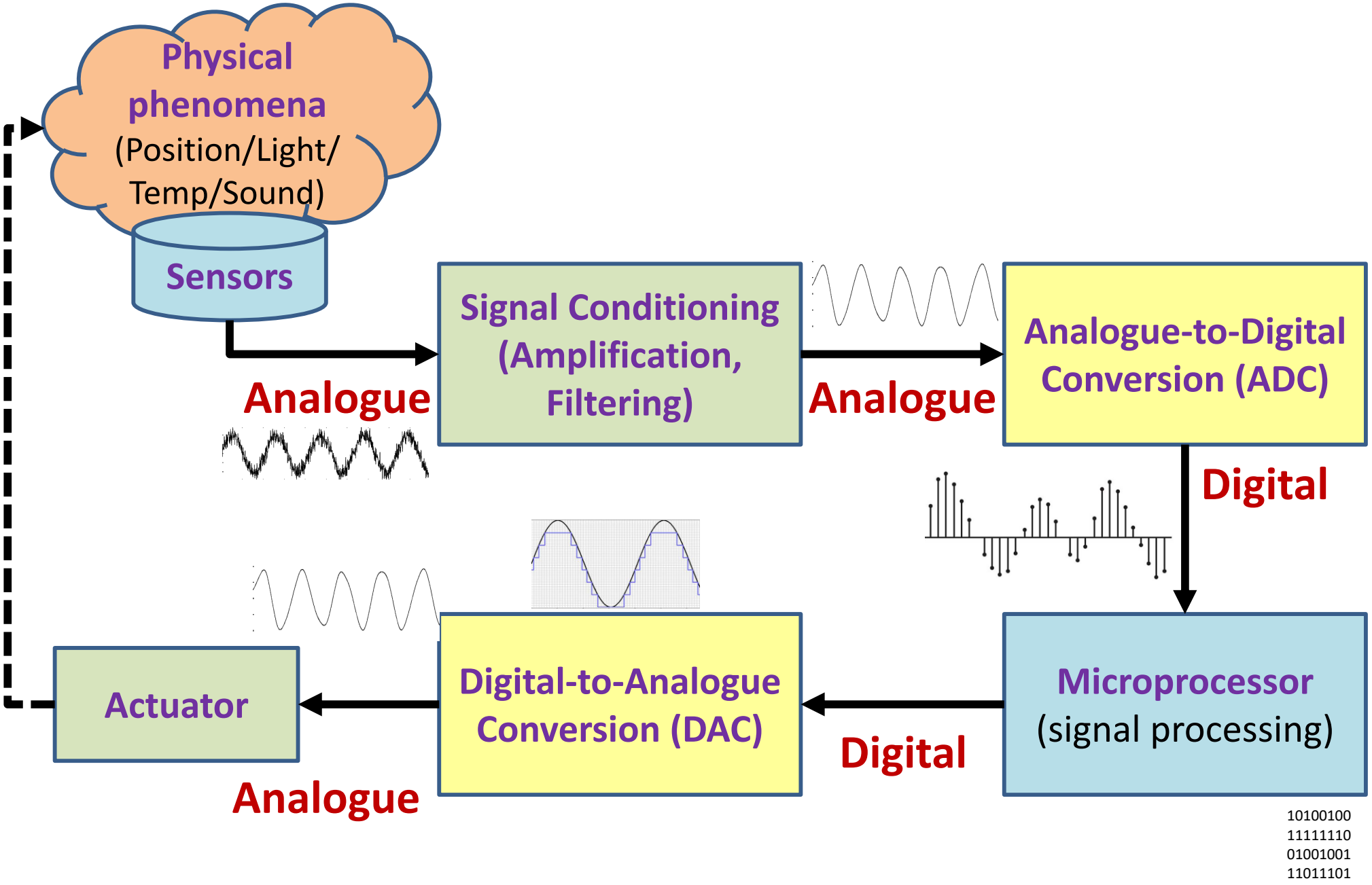


EVALBOT from Texas Instruments

The Measurement and Control Loop

- The quality of analogue signals typically needs to be improved before they are converted into digital signals by the system's data acquisition hardware
  - Signal conditioning (Noise filtering, amplification, attenuation)
- The signals that have been conditioned are converted into digital formats suitable for microprocessor to process (e.g. digital signal processing algorithms)
  - Analogue-to-Digital Converter (ADC)
- Processed digital signals are changed back to analogue signals so that they can be recognized by human senses (e.g. audio, video signals) or used to control actuators
  - Digital-to-Analogue Converter (DAC)

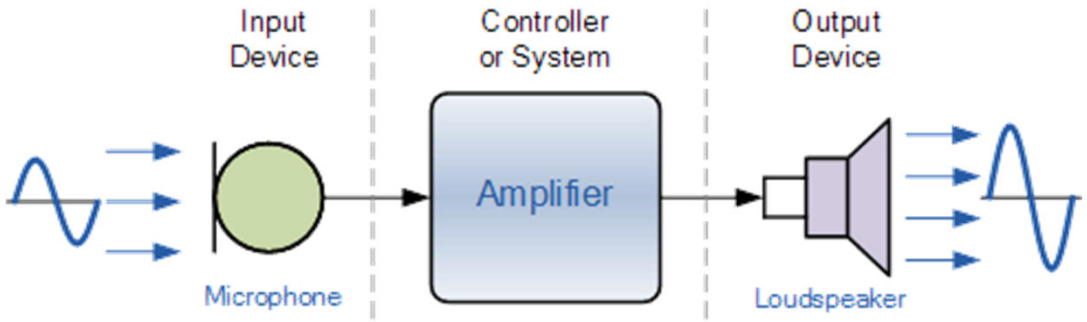
The Measurement and Control Loop



Sensors and Actuators

Sensors and Actuators

- Sensors
  - Acquires information from the real world
  - Device that responds to physical stimulus (**measurand**) such as light, heat, sound, etc.
  - A transducer is often needed to convert the measurand to other forms of energy (e.g. electrical signals)
- Actuator
  - Device or mechanism capable of converting a controller command signal into a change in physical parameters (e.g. loudspeakers, motors)



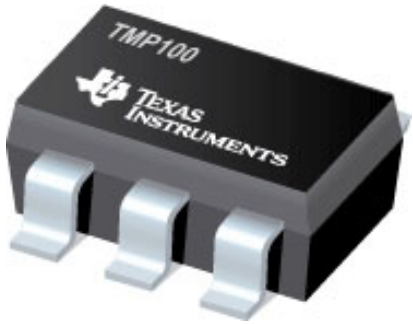
Examples of Sensors and Actuators

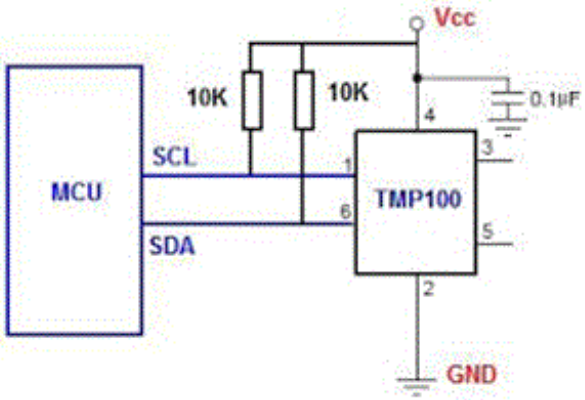
Measurand	Input Device (Sensor)	Output Device (Actuator)
Light Level	Light Dependant Resistor (LDR) Photodiode Photo-transistor Solar Cell	Lights & Lamps LED's & Displays Fibre Optics
Temperature	Thermocouple Thermistor Thermostat Resistive temperature detectors (RTD)	Heater Fan
Force/Pressure	Strain Gauge Pressure Switch Load Cells	Lifts & Jacks Electromagnet Vibration
Position	Potentiometer Encoders Reflective/Slotted Opto-switch LVDT	Motor Solenoid Panel Meters
Speed	Tacho-generator Reflective/Slotted Opto-coupler Doppler Effect Sensors	AC and DC Motors Stepper Motor Brake
Sound	Carbon Microphone Piezo-electric Crystal	Bell Buzzer Loudspeaker

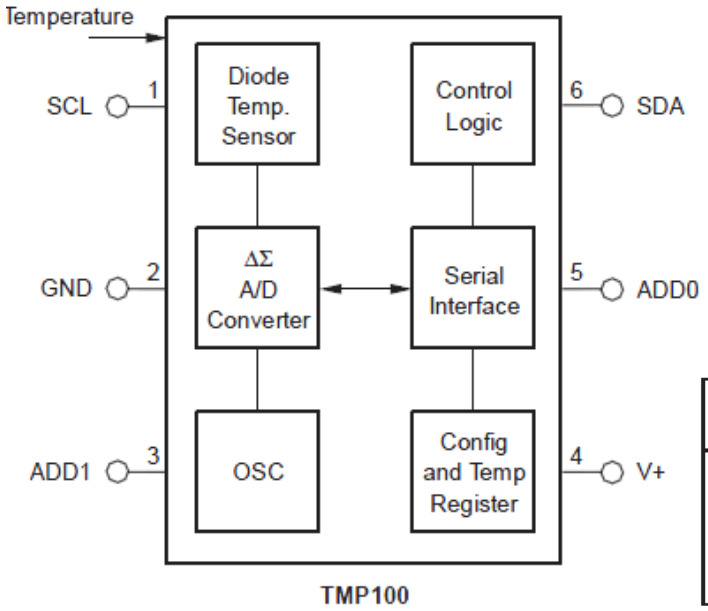
Sensors and Transducers: Source: http://www.electronics-tutorials.ws/io/io_1.html

Interfacing Temperature Sensor Using ADC

- TMP100 Temperature Sensor (Used in Lab 3)
  - Temperature range: -55°C to 125°C
  - Temperature resolution of up to 0.0625 °C
  - Voltage range: 2.7V to 5.5V
  - Communication protocol: I2C







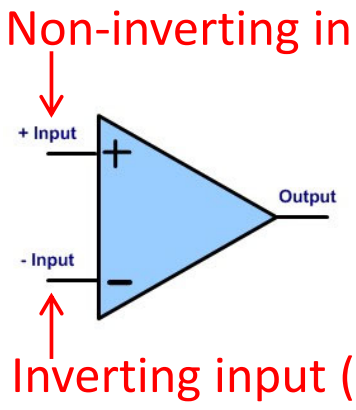
TEMPERATURE (°C)	DIGITAL OUTPUT (BINARY)	HEX
128	0111 1111 1111	7FF
127.9375	0111 1111 1111	7FF
100	0110 0100 0000	640
80	0101 0000 0000	500
75	0100 1011 0000	4B0
50	0011 0010 0000	320
25	0001 1001 0000	190
0.25	0000 0000 0100	004
0.0	0000 0000 0000	000
−0.25	1111 1111 1100	FFC
−25	1110 0111 0000	E70
−55	1100 1001 0000	C90
−128	1000 0000 0000	800

R1	R0	RESOLUTION	CONVERSION TIME (typical)
0	0	9 Bits (0.5°C)	40ms
0	1	10 Bits (0.25°C)	80ms
1	0	11 Bits (0.125°C)	160ms
1	1	12 Bits (0.0625°C)	320ms

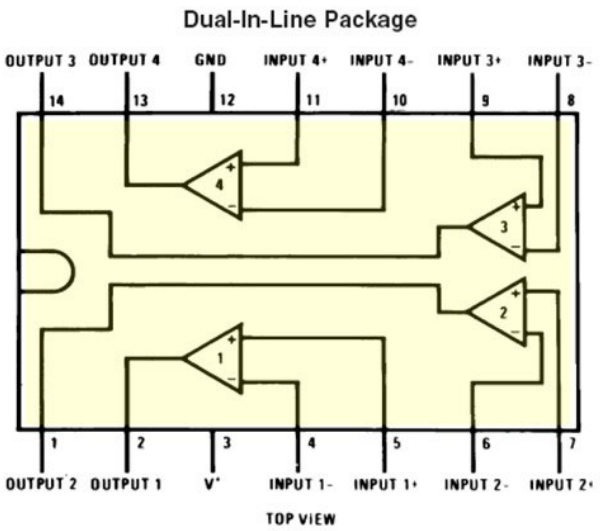
Source: http://www.ti.com/lit/ds/sbos231g/sbos231g.pdf

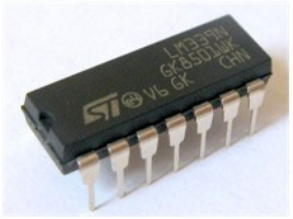
Voltage Comparator

- If (+) input is above its (-) input, output is pulled High
- If (+) input is below its (-) input, output is pulled Low



http://www.ermicro.com/blog



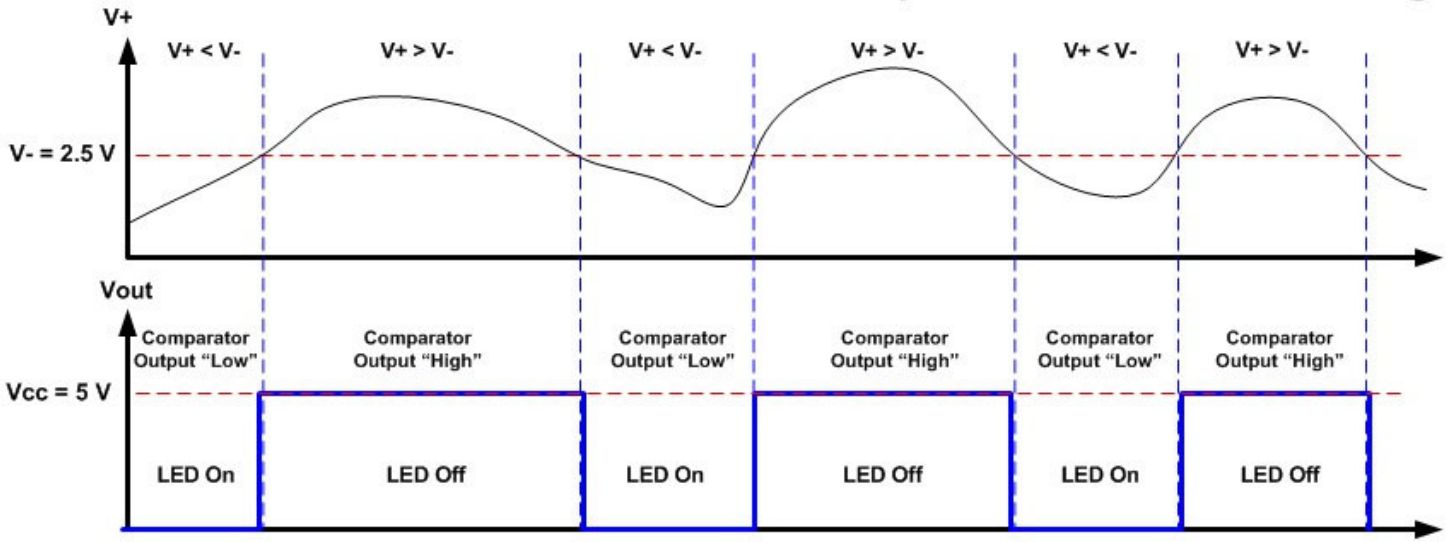


LM339 Quad IC

LM339 Quad Compartor

National Semiconductor LM339 Quad Comprator, 4 Independent Comparator in 14 Pins DIL Package

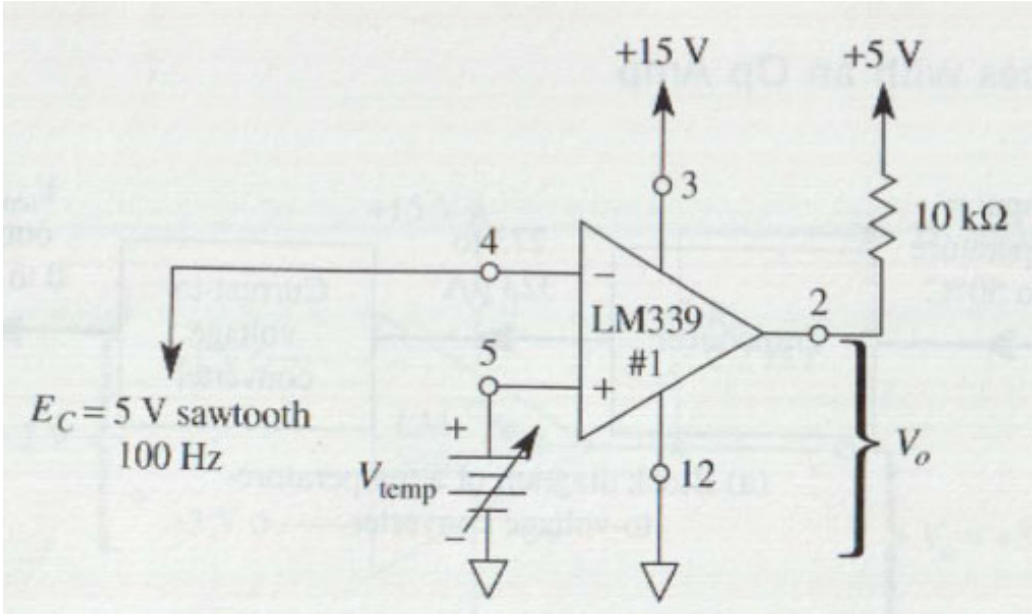
http://www.ermicro.com/blog



LM339 Single Comparator Testing Circuit Graph

Non-Inverting Pulse Width Modulator

- Measured voltage (V_{temp}) is connected to non-inverting input (+) of voltage comparator
- Sawtooth wave (E_c) with constant frequency is connected to (-)
- Rate of change of V_{temp} must be less than E_c
- Output is defined as the time, T_H, that V_O stays high

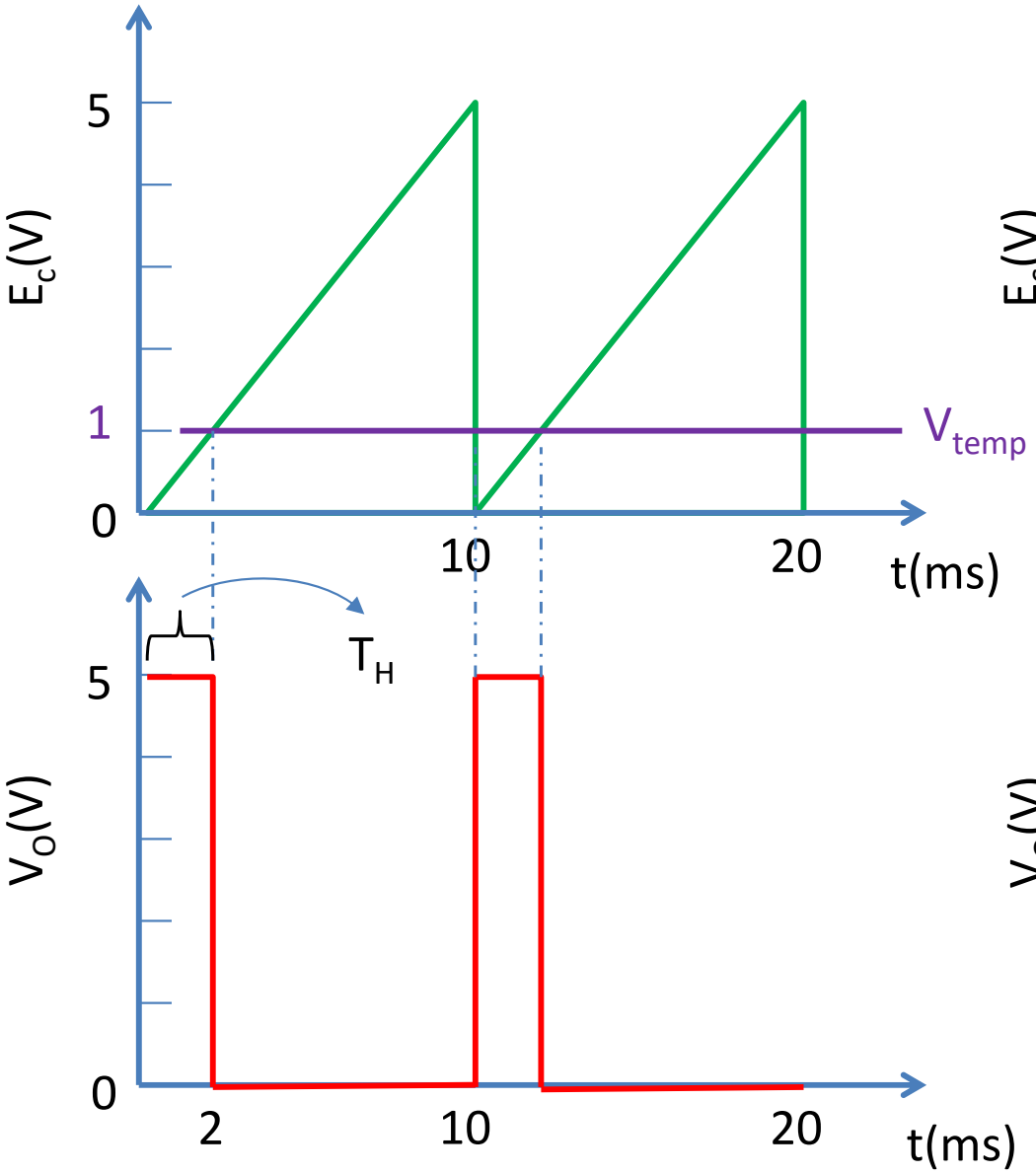


Source: Coughlin and Driscoll, "Operational Amplifiers and Linear Integrated Circuits", Prentice Hall, 6th Edition

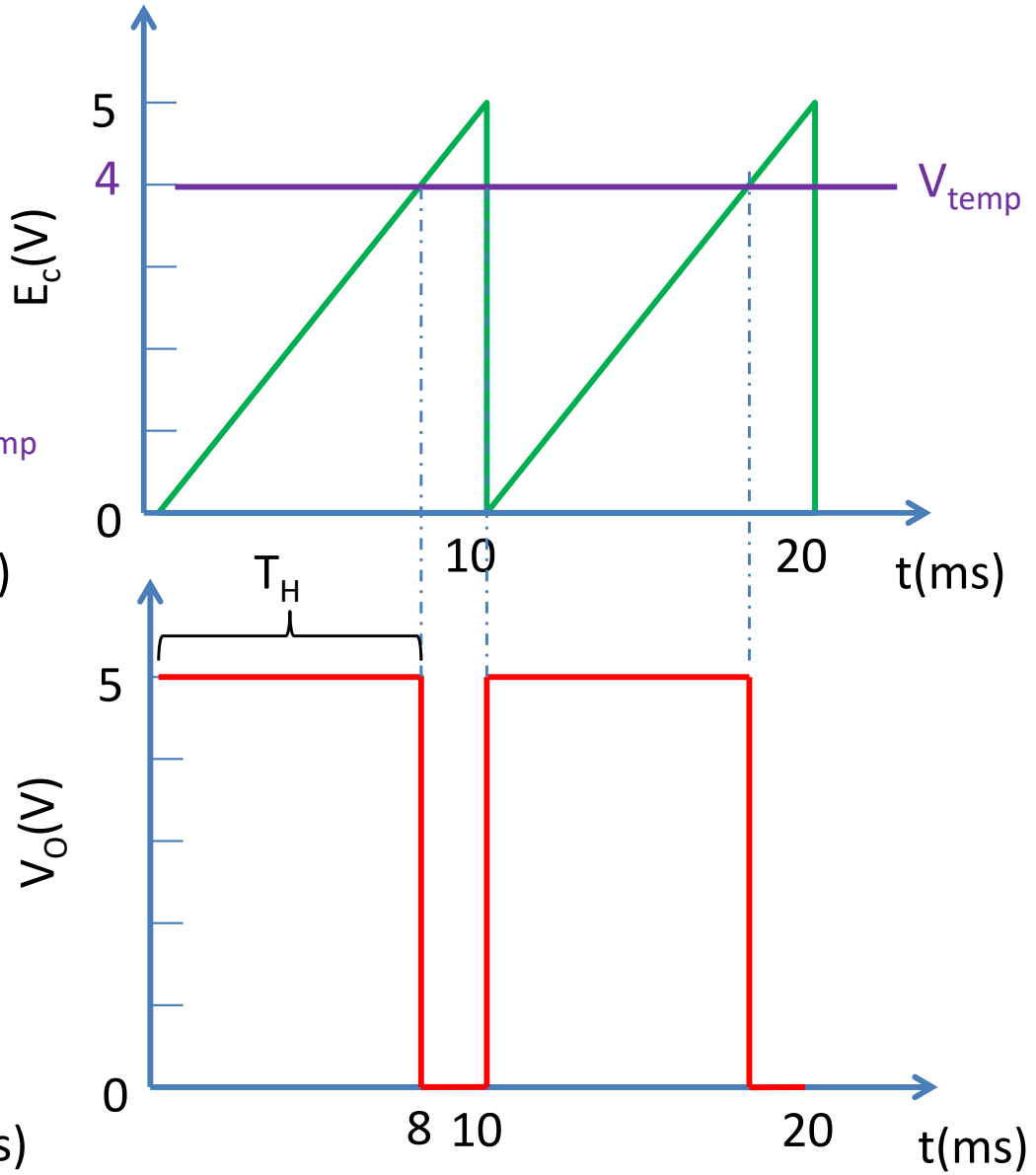
Non-Inverting Pulse Width Modulator (Cont.)

Back

- When V_{temp} = 1V, output stays high for 2ms

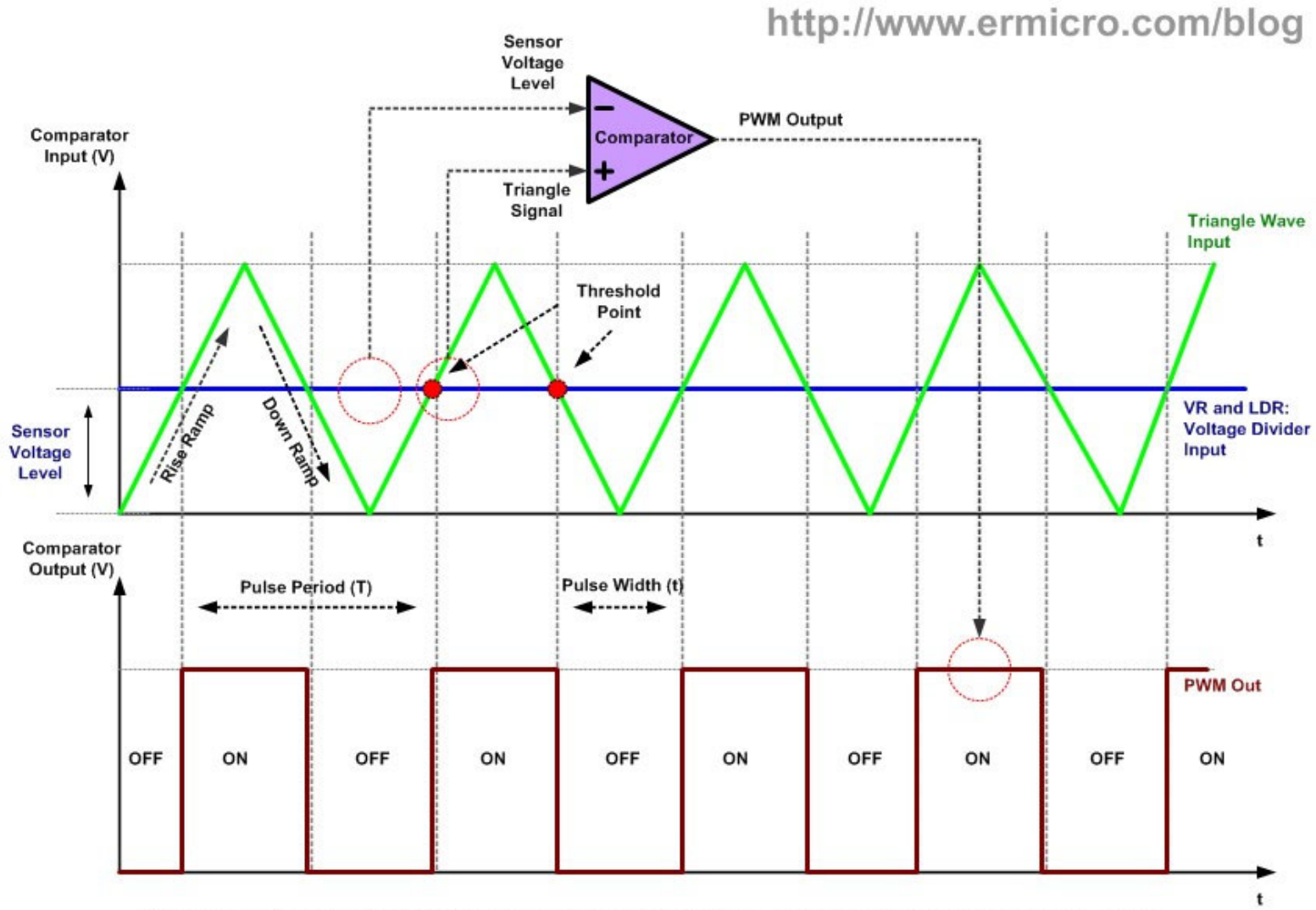


- When V_{temp} = 4V, output stays high for 8ms



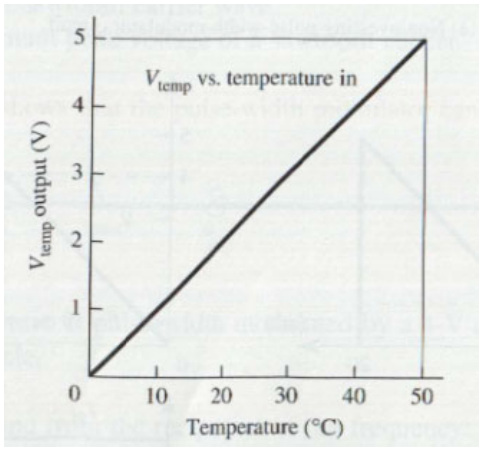
Interfacing Analogue Sensor Using Voltage Comparator to Generate PWM

- This is an inverting PWM as the measured sensor voltage is connected to inverting input (-) of voltage comparator



Voltage Control PWM Using Triangle Wave and Variable Voltage Level

Interfacing Temperature Sensor with Microprocessor



Input-output characteristic of sensor (from datasheet)

Temperature 0 to 50°C

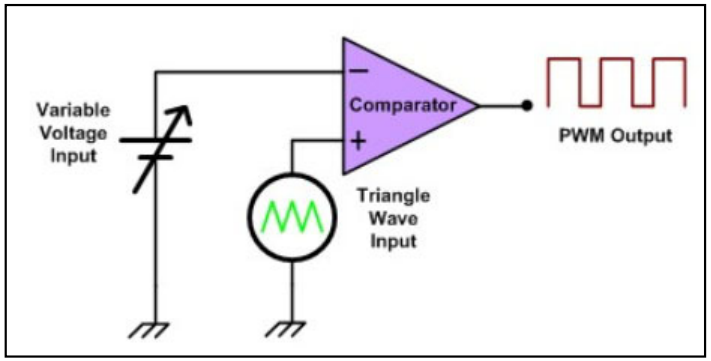
Temperature Sensor

V_{temp} 0 to 5V

LM339

Required temperature

Microprocessor T_H

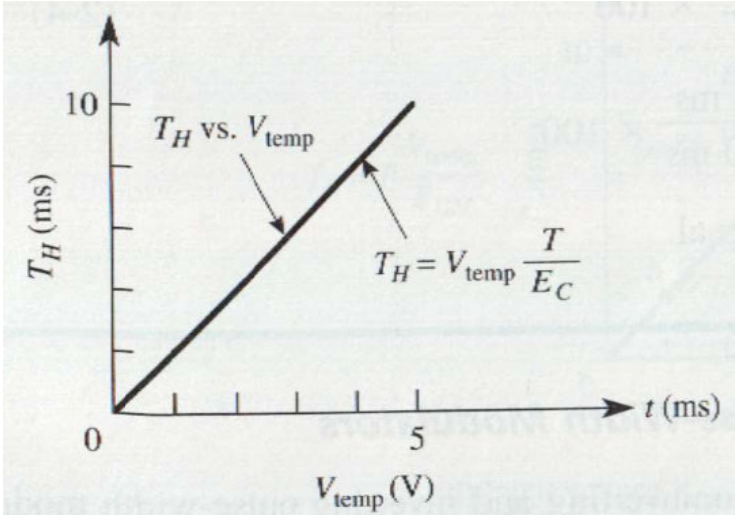


Control the room temperature

- Use a timer to determine duration of T_H

Transfer Function

$V_{temp} = (T_H * E_c)/T$



Input-output characteristic of voltage comparator

```
void main(void)
   {
   /* Init the system */
   C_HEAT_Init();

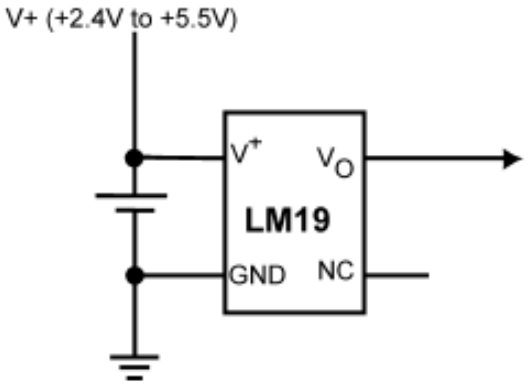
   while(1) /* 'for ever' (Super Loop) */
      {
      /* Find out what temperature the user requires
         (via the user interface) */
      C_HEAT_Get_Required_Temperature();

      /* Find out what the current room temperature is
         (via temperature sensor) */
      C_HEAT_Get_Actual_Temperature();

      /* Adjust the gas burner, as required */
      C_HEAT_Control_Boiler();
      }
   }
```

Transfer Function

- The measured temperature can be calculated using a transfer function



20004002

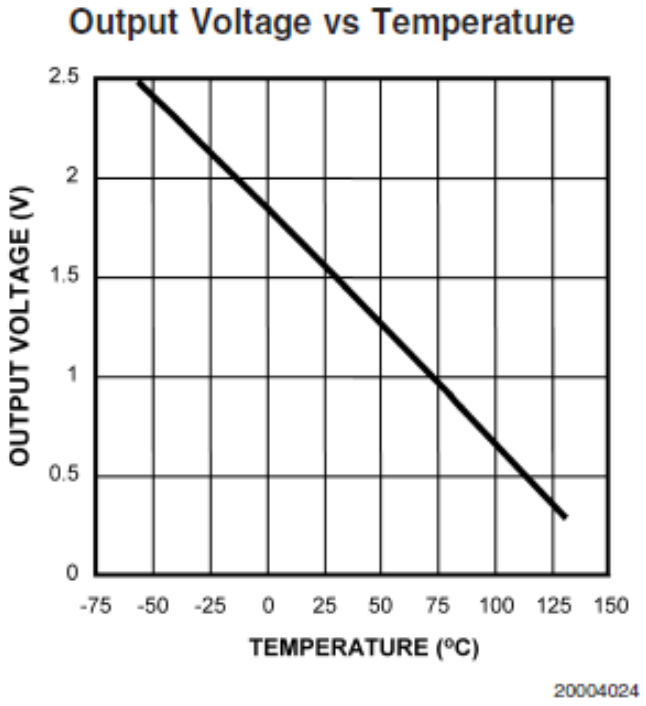
$$V_O = (-3.88 \times 10^{-6} \times T^2) + (-1.15 \times 10^{-2} \times T) + 1.8639$$

or

$$T = -1481.96 + \sqrt{2.1962 \times 10^6 + \frac{(1.8639 - V_O)}{3.88 \times 10^{-6}}}$$

where:

T is temperature, and V_O is the measured output voltage of the LM19.

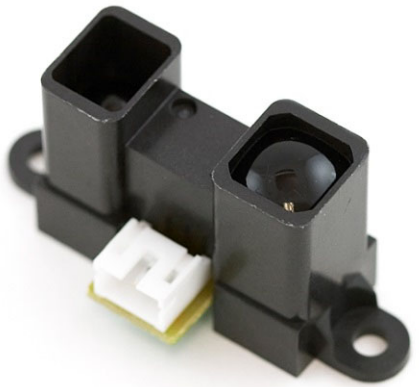


Source: National Semiconductor LM19 TO-92 Temperature Sensor

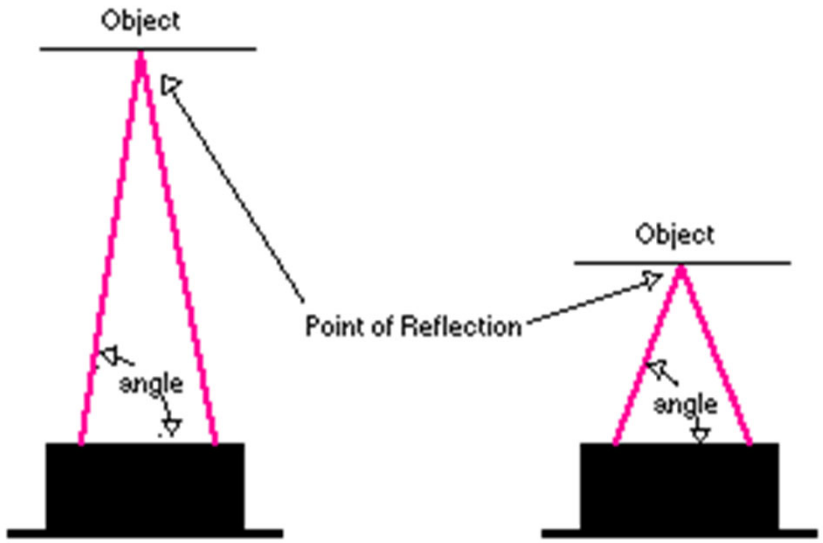
- You may need to derive the transfer function if you are designing the signal conditioning circuit

Infrared Proximity Sensor

- Typically used for distance and object detection
- Based on the concept of **triangulation**
  - Pulse of light is emitted and reflected by object
  - The light returns at an angle that is dependent on the distance of the object



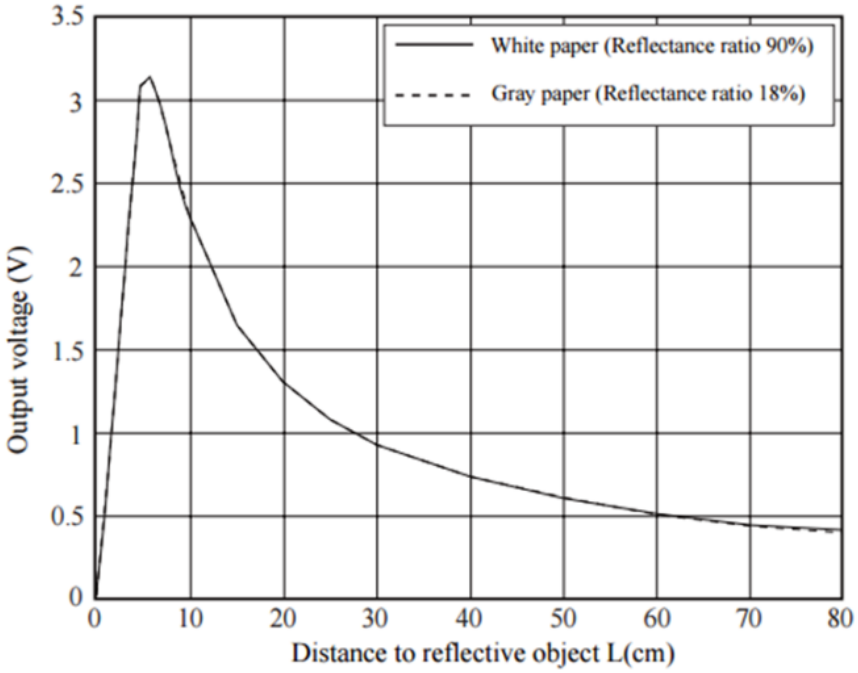
Source: https://www.sparkfun.com/products/8958



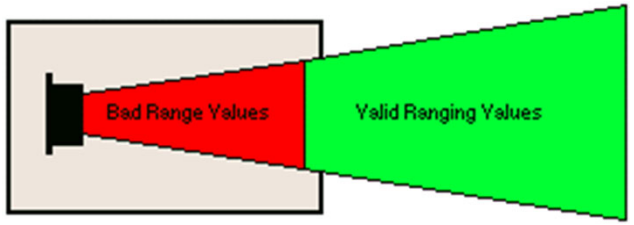
Source: http://www.societyofrobots.com/sensors_sharpirrange.shtml

Infrared Proximity Sensor (Cont.)

- Problems with IR sensor
  - Transfer function provided by datasheet may not be accurate
  - Non-Linear output
    - Transfer function (e.g. V = K(1/L)) requires a division
  - Signals are noisy
  - Cannot get accurate reading for close objects



- Possible Solutions
  - Gather sensor data to obtain transfer function
  - Use a lookup table to store voltage/distance pairs
  - Take multiple samples and find the mode (most frequent value)
  - Attach the sensor such that the front of the robot is located before the minimum sensor range



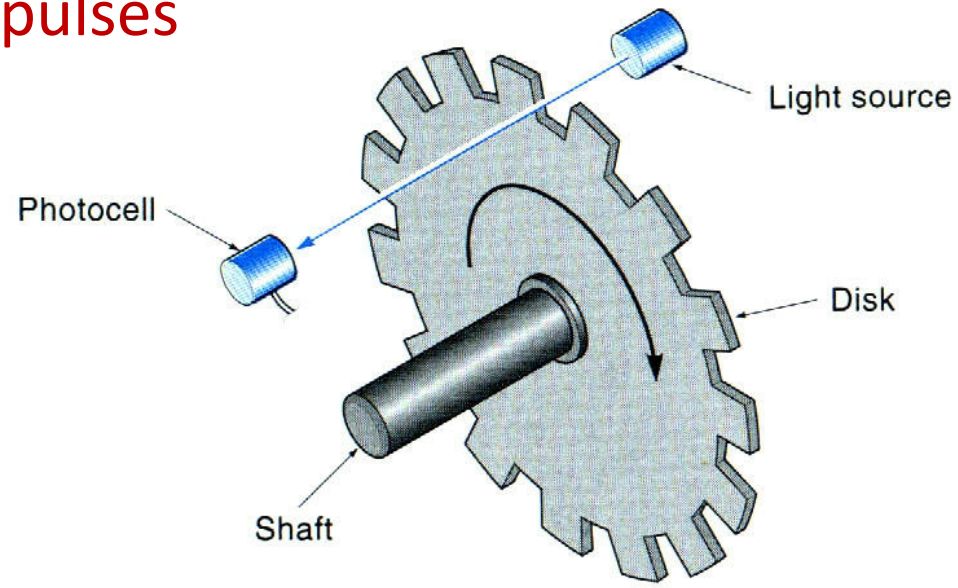
Source: http://www.societyofrobots.com/sensors_sharpirrange.shtml

Position Sensors

- Measure the position and/or displacement of physical objects
  - With reference from a fixed point or position
- Linear Sensors
  - Detect movement of an object in a **straight line**
- Rotational Sensors
  - Detect movement of an object by its **angular movement**
- Examples
  - Potentiometer
  - Rotary Encoders

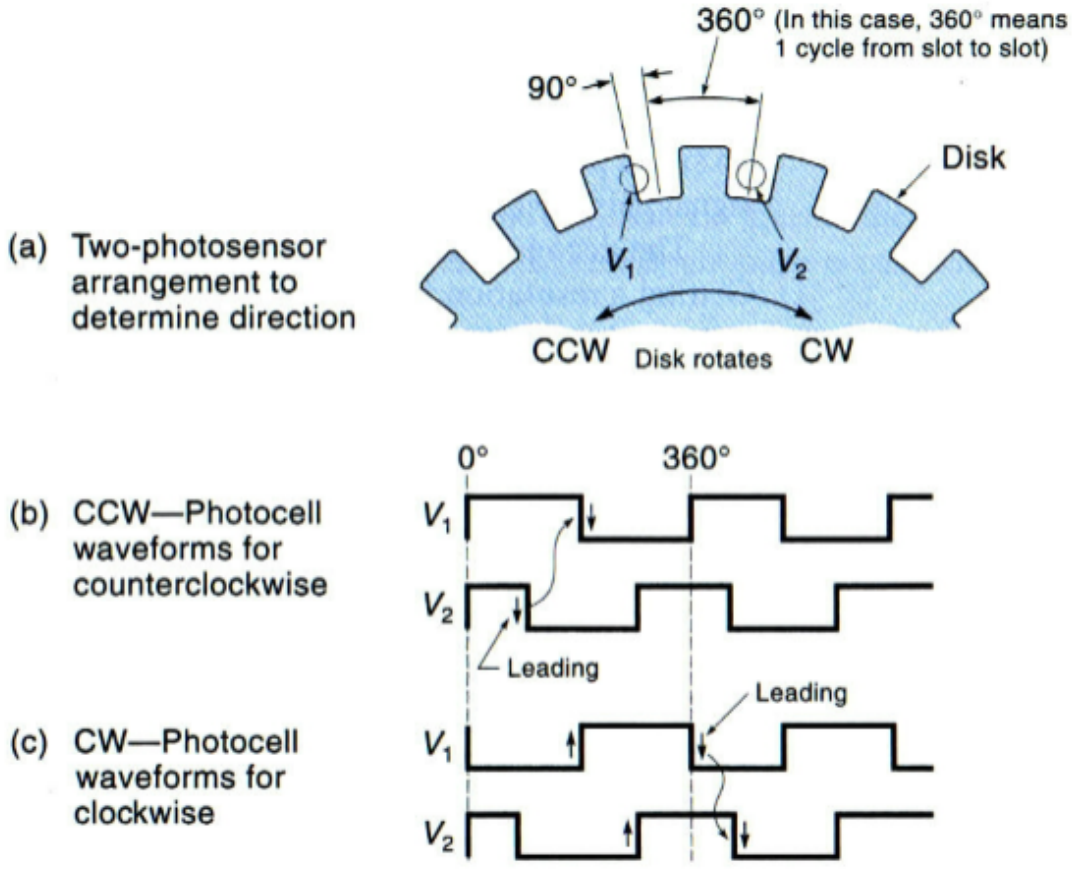
Rotary Encoders

- Non-contact optical devices used for converting the angular position of a rotating shaft into a digital data code
- LED or infra-red light source is passed through a rotating encoded disk and photo detectors scan the disk as it rotates
  - Electronic circuit processes the information into a digital form as a stream of binary output pulses
- Two basic types of rotary encoders
  - Absolute Position Encoder
  - Incremental Encoder (Quadrature encoders)

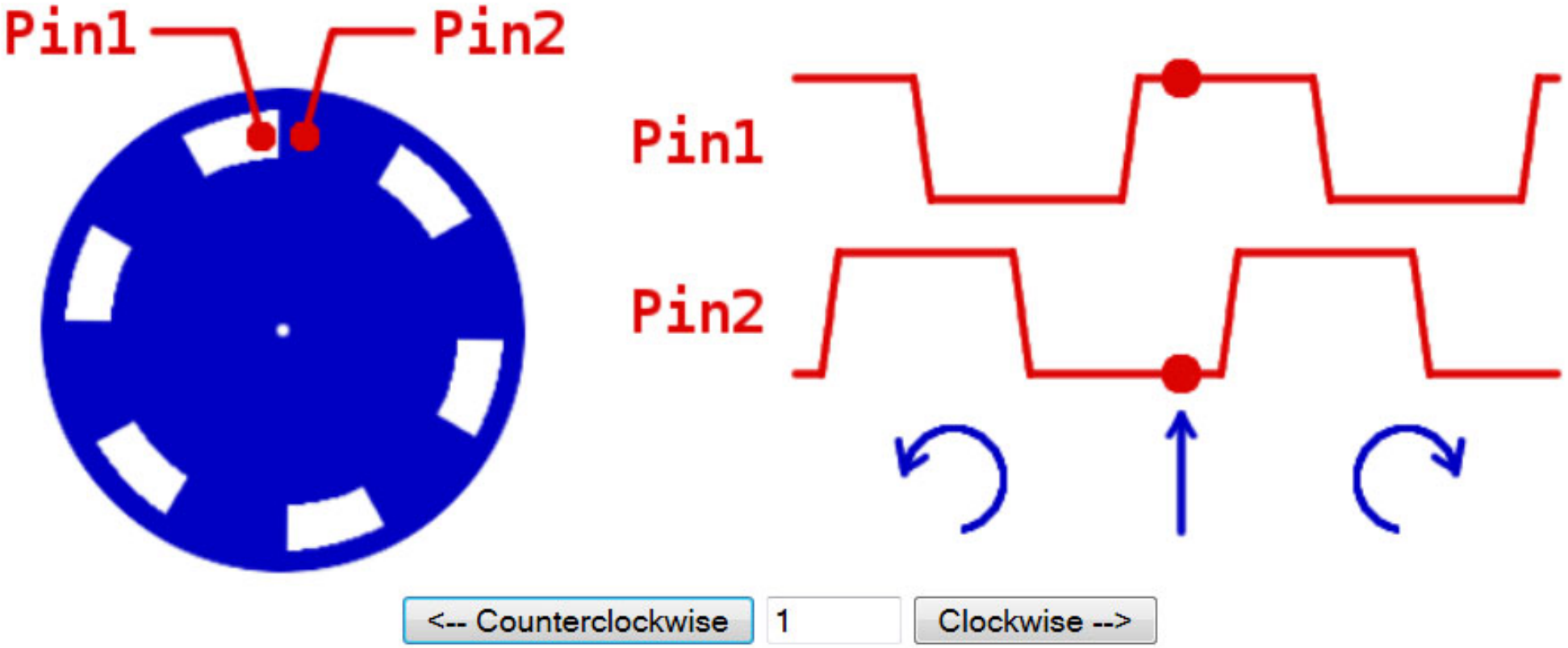


Incremental Encoders

- Require a coded disk, with evenly spaced transparent and dark lines
- Produces a stream of square wave pulses which can be used to determine movement amount and direction
- By monitoring outputs V_1 and V_2 (90° out of phase), we can determine:
  - Direction of rotation
  - How far it has turned (by counting the number of pulses)
  - How fast it is being turned (by counting the frequency of the pulses)



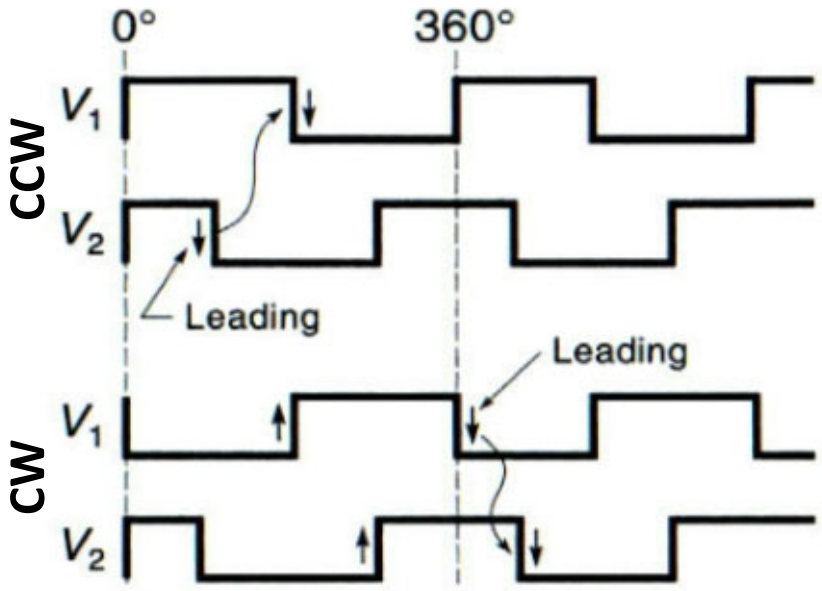
Incremental Encoders

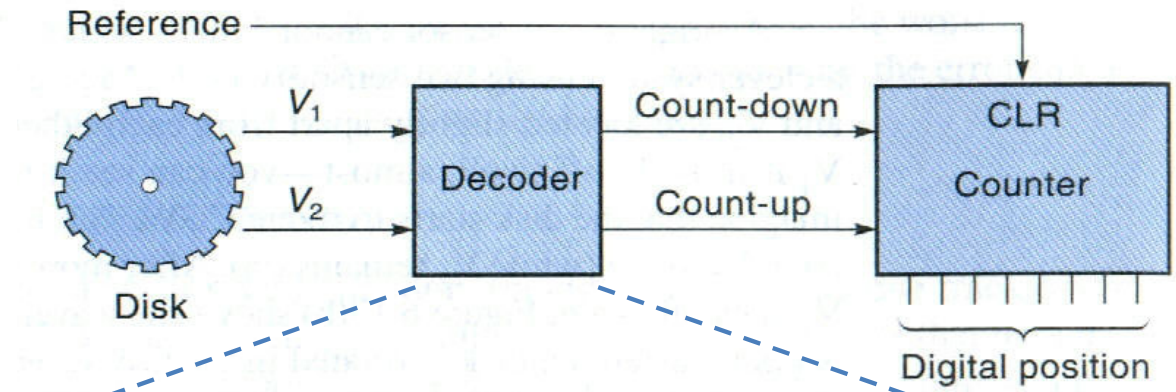


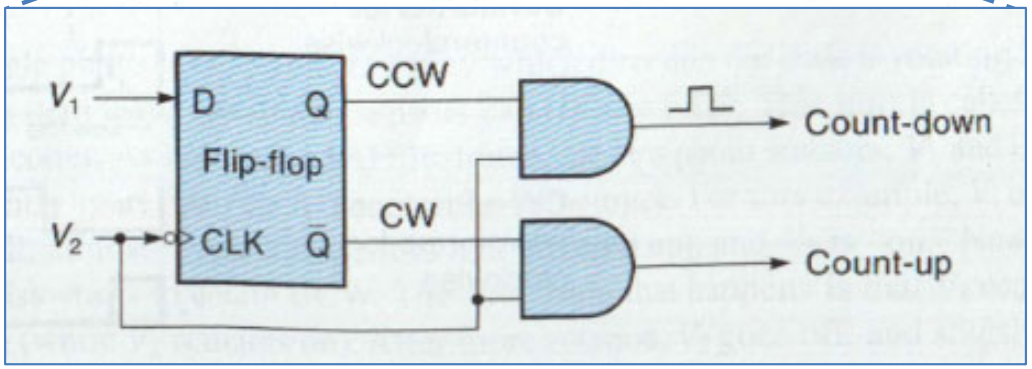
Source: http://www.pjrc.com/teensy/td_libs_Encoder.html

Determine Direction of Rotation and Number of Turns

- When disc is going counter-clockwise (CCW)
  - V_2 leads V_1 by 90° (Every time V_2 goes low, V_1 is high)
  - Output of FF is high and count-down pulse is generated
- When disc is going clockwise (CW)
  - V_1 leads V_2 by 90° (Every time V_2 goes low, V_1 is low)
  - Output of FF is low and count-up pulse is generated

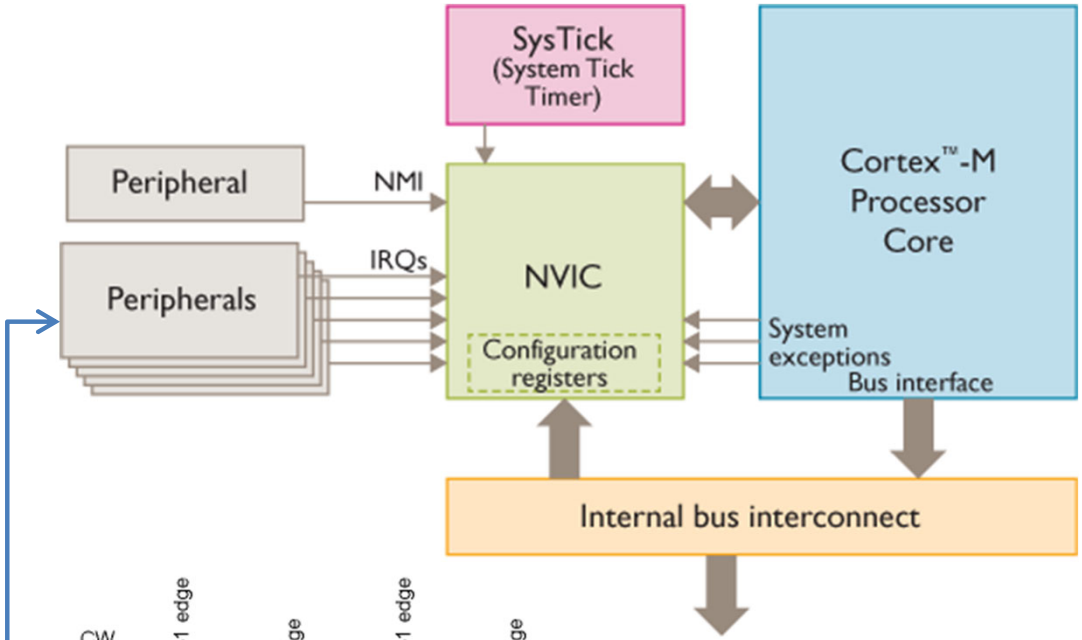


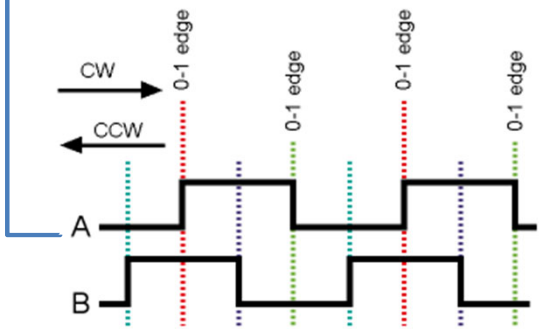


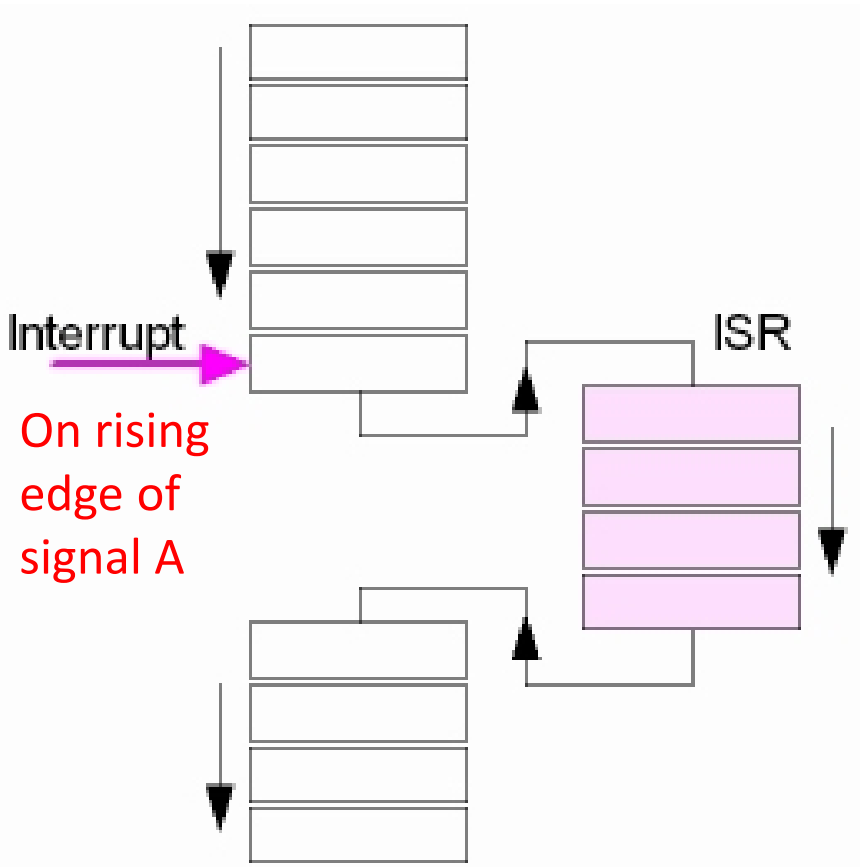


Interfacing Incremental Encoders (Using Interrupt)

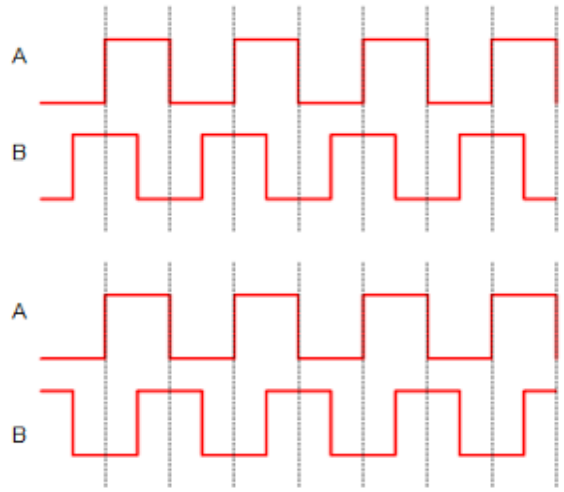
- Use interrupt to detect rising edge of signal A
- Write an interrupt service routine (ISR) to increment/decrement counter
  - ISR is a block of code that the microcontroller will jump to when the interrupt is triggered

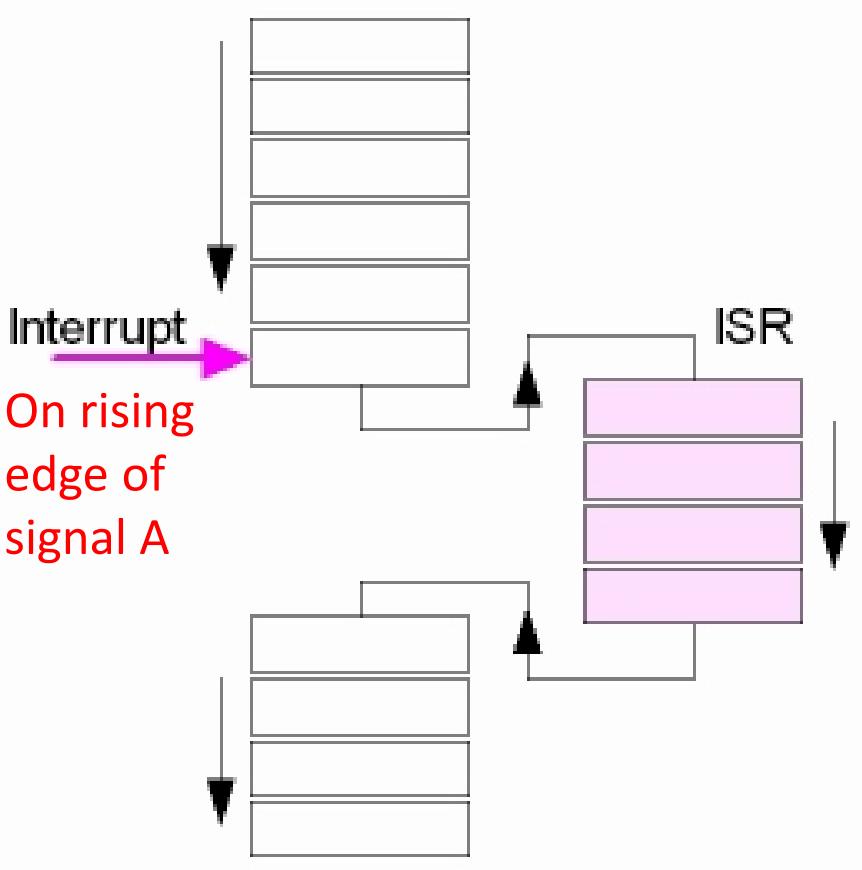


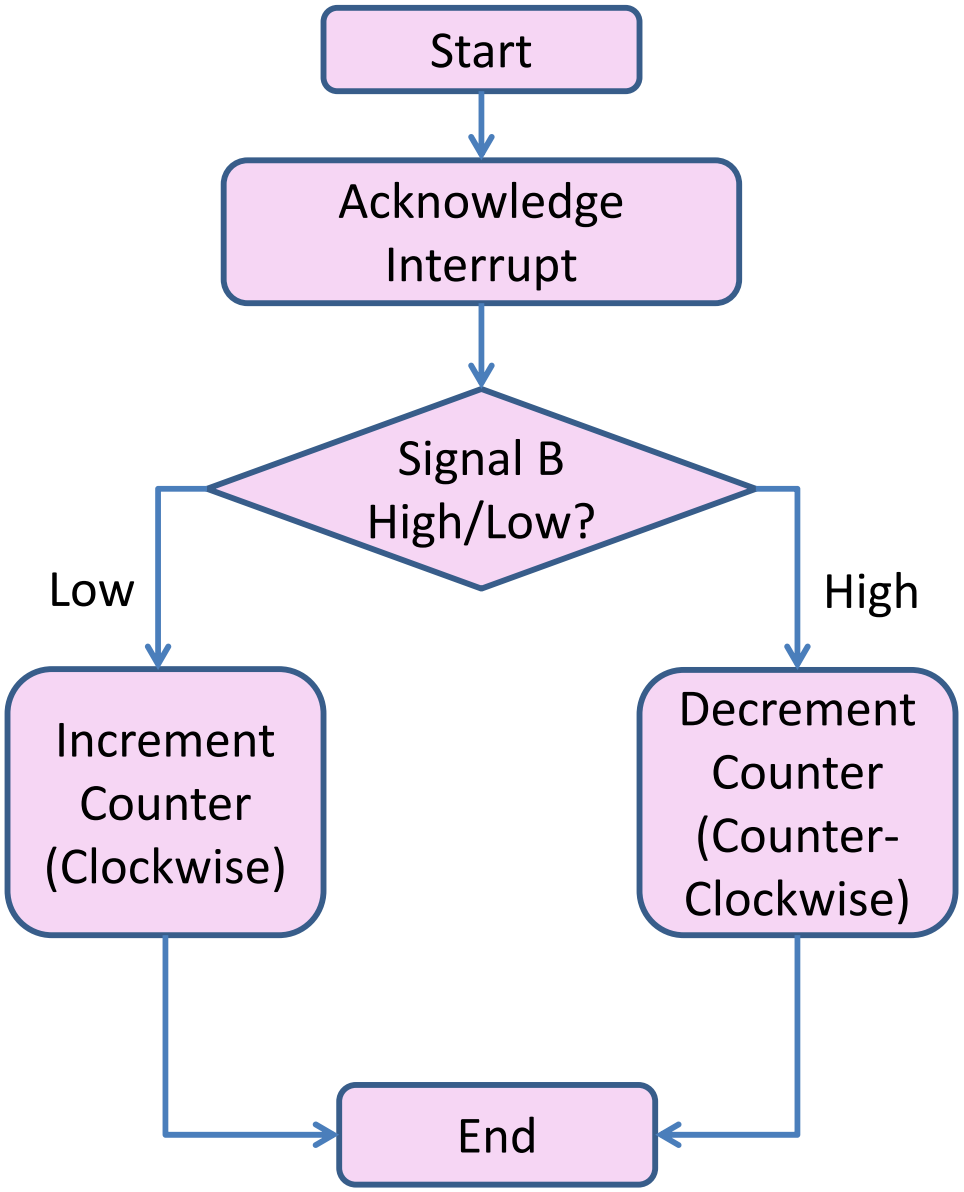




Interfacing Incremental Rotary Encoder (Interrupt)





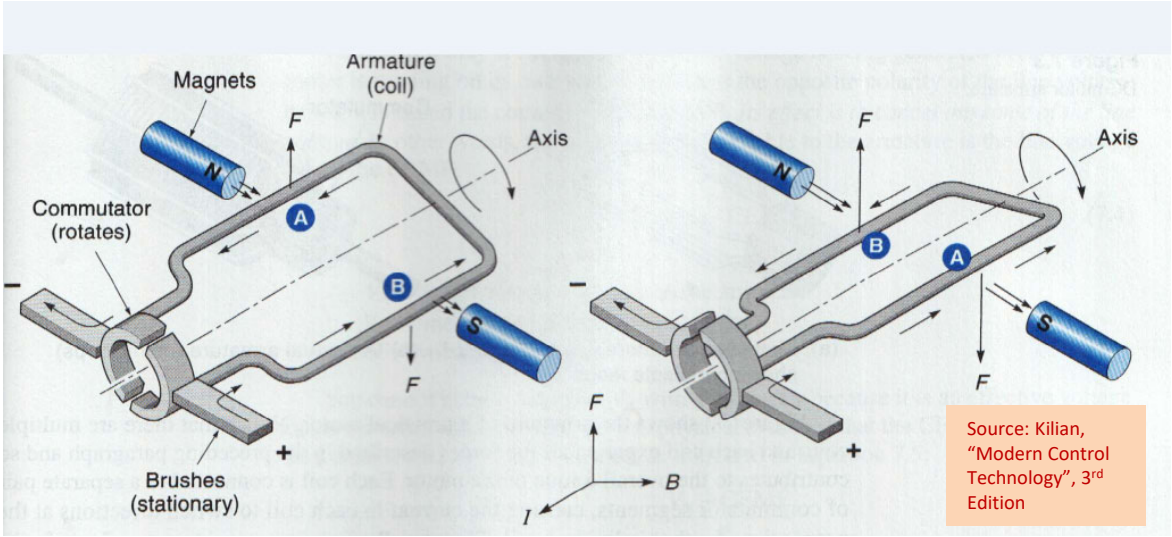


Interrupt Service Routine (ISR)

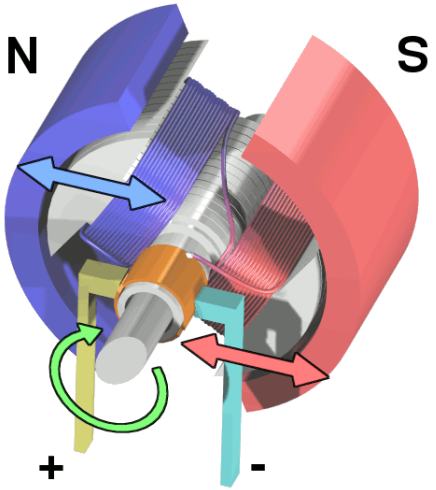
DC Motors

- Device which converts electrical energy to mechanical energy

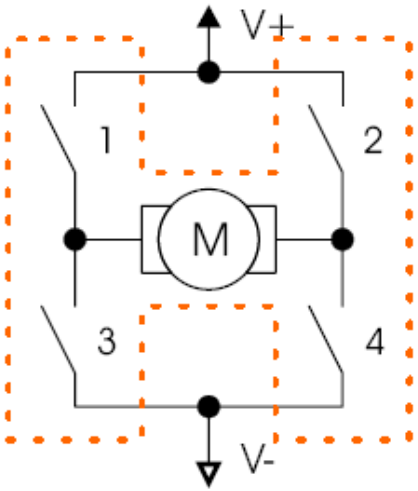


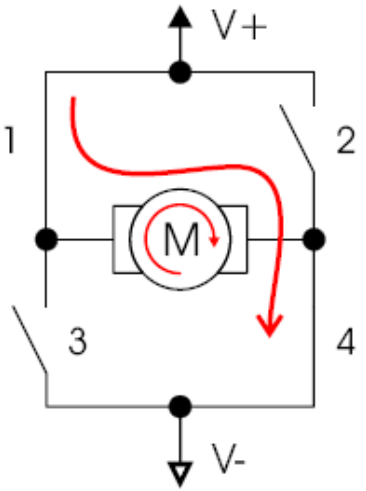


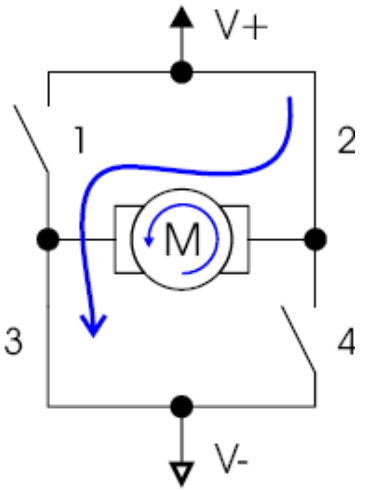
Source: Kilian, "Modern Control Technology", 3rd Edition



- The basic circuit for driving the DC motor in both directions is an H-Bridge

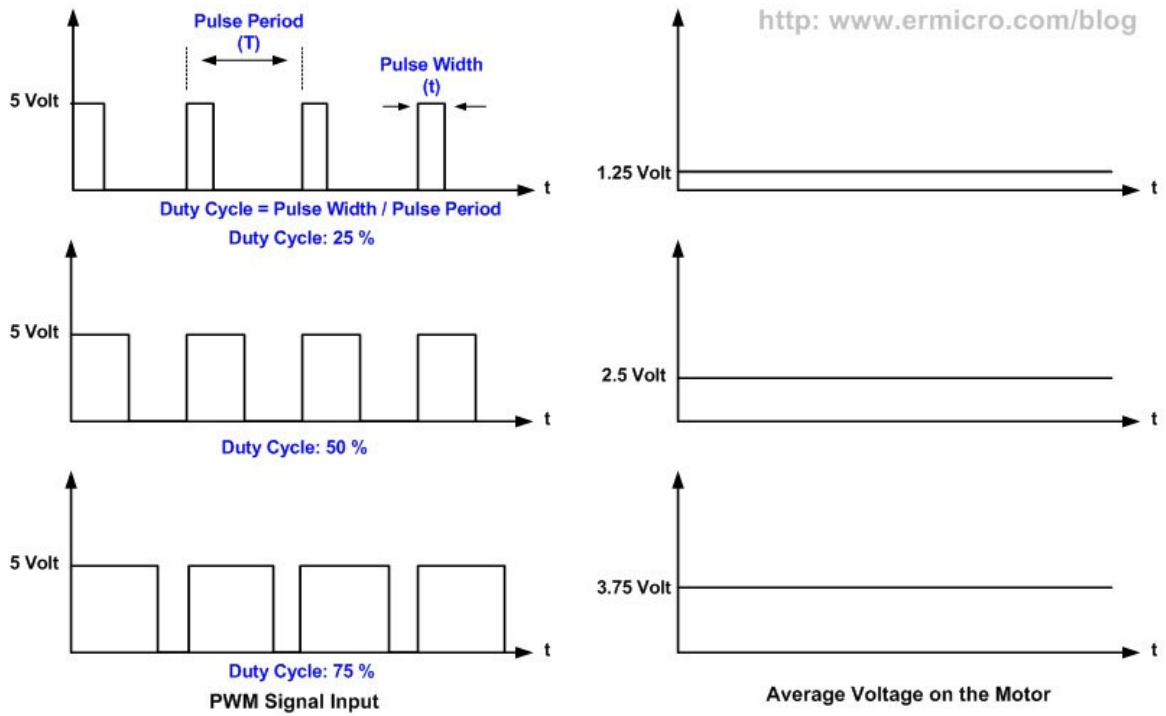






Pulse Width Modulation

- Pulse width modulation is a technique for reducing the amount of power delivered to a DC motor
  - This can then be used to control the speed of the motor
- Instead of reducing the voltage operating the motor (which would reduce its power), the motor's power supply is rapidly switched on and off

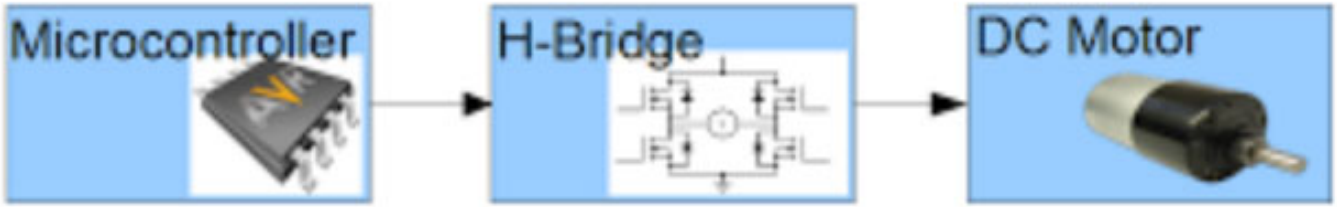


PWM Timing Diagram

Motor Control

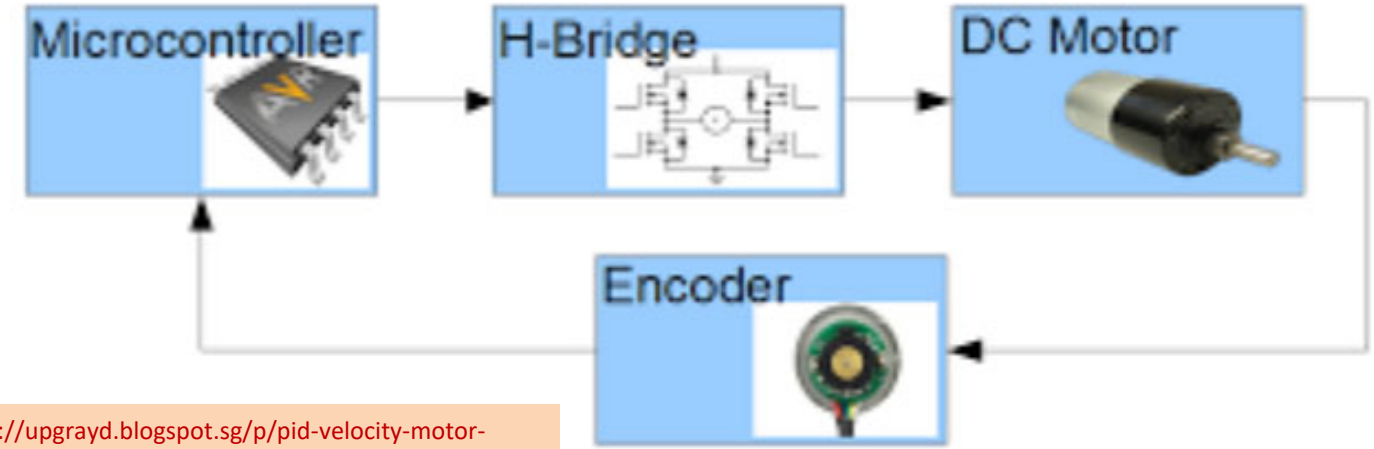
- Microcontroller provides H-Bridge with direction and speed information. H-Bridge drives the DC motor.

Open Loop Motor Control



- However, feedback is needed if we want to produce accurate speed and movement
- The encoders outputs can be decoded by the microcontroller to calculate the rotational velocity of the motor

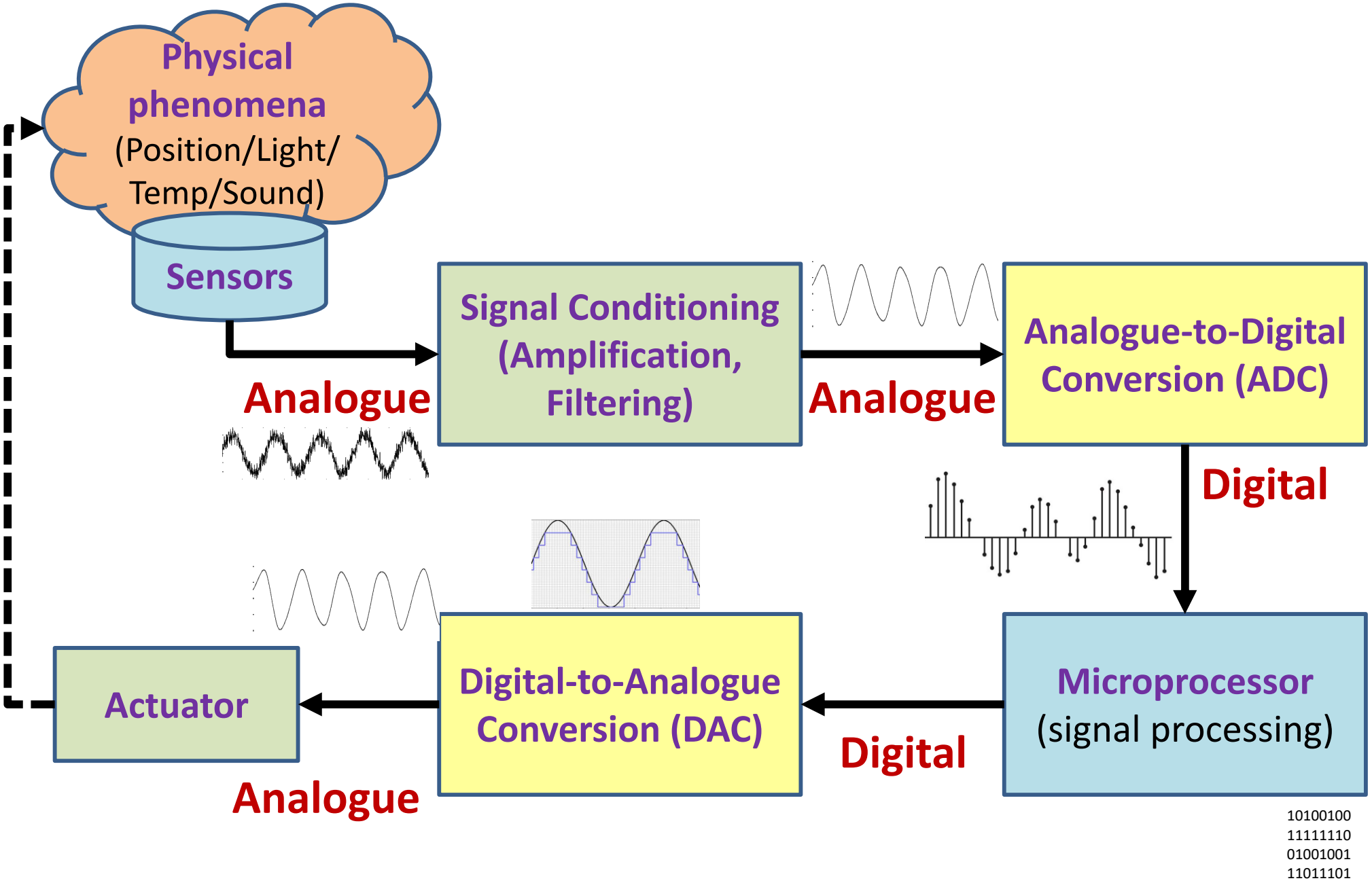
Closed Loop Motor Control



Source: http://upgrayd.blogspot.sg/p/pid-velocity-motor-controller.html

Signal Conditioning Circuits

The Measurement and Control Loop



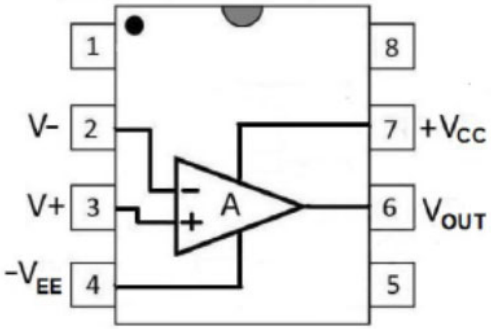
Why We Need to Amplify Signal?

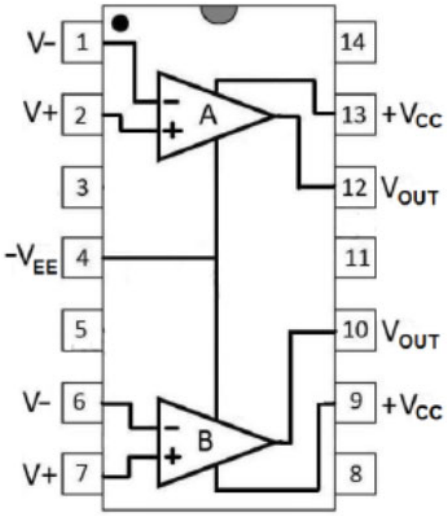
- For maximum resolution, the voltage range of the input signals should be approximately equal to the maximum input range of the A/D converter
- Amplification expands the range of the transducer signals so that they match the input range of the A/D converter
  - For example, a 10x amplifier maps transducer signals which range from 0 to 1 V into the range 0 to 10 V before they go into the A/D converter
- Amplifiers are used to 'condition' the analog signals to make them suitable for the ADC
  - If the signal is too small, it can be amplified
  - If the signal is too big, it can be attenuated

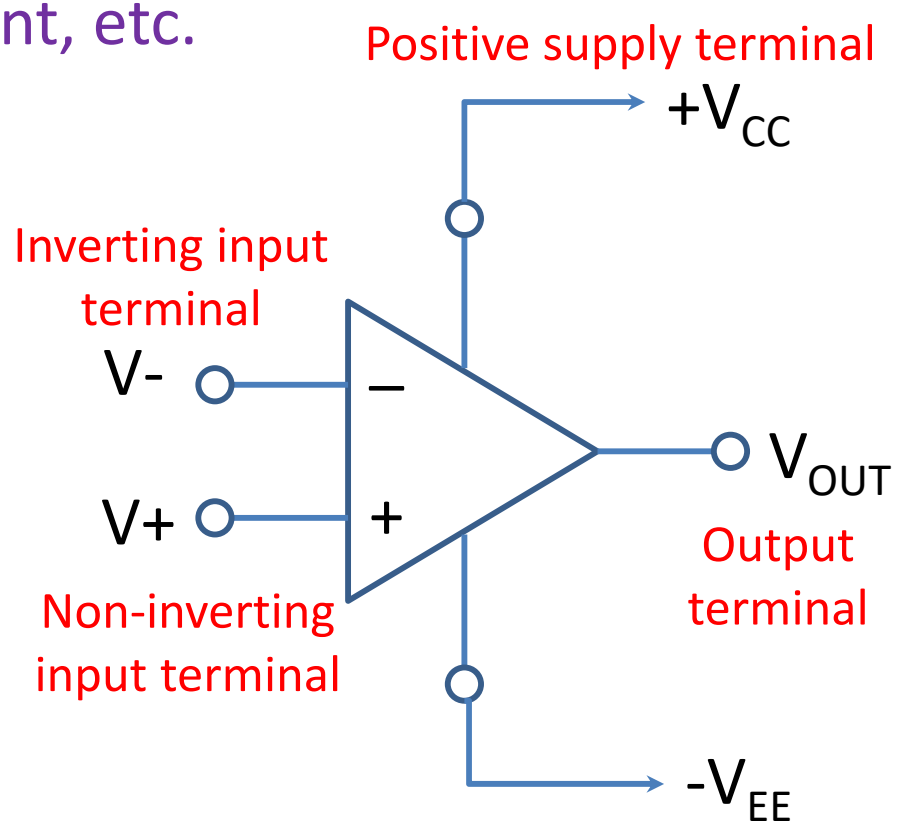
Operational Amplifier

- Operational amplifier (Op-amp) is one of the most versatile and widely-used integrated circuit in electronics industry
  - Inexpensive and dependable
  - Signal conditioning, process control, communications, power and signal sources, measurement, etc.





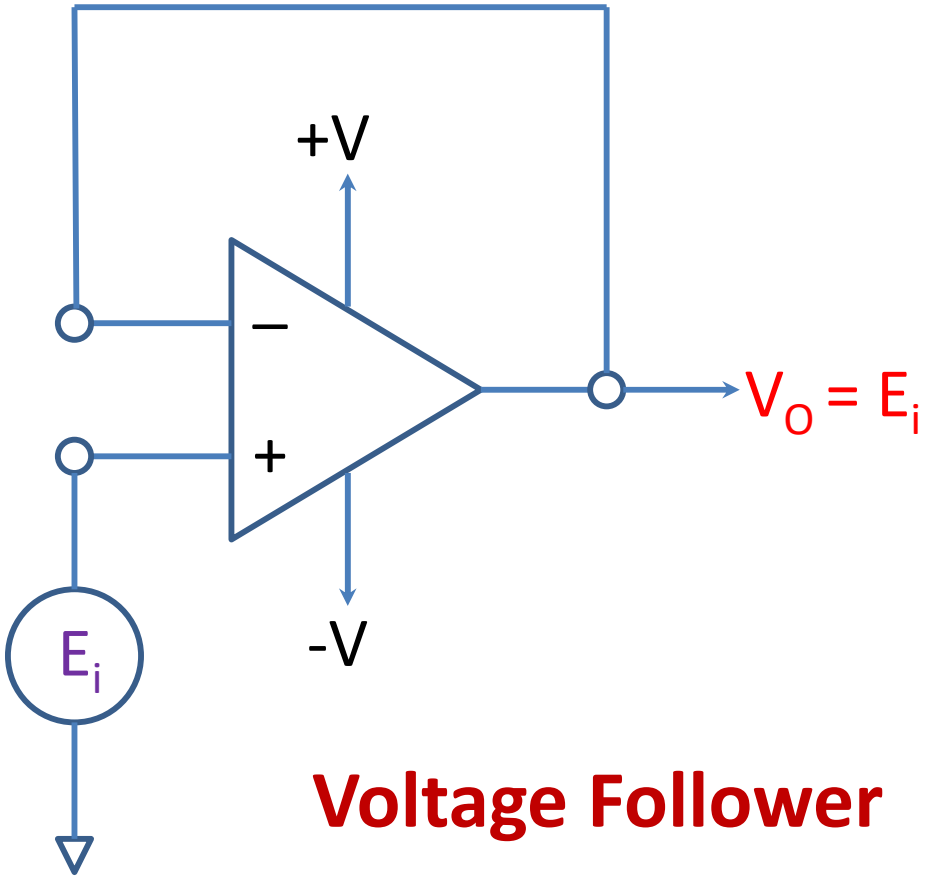




Circuit symbol for general purpose op-amp

Op-Amp as Amplifier

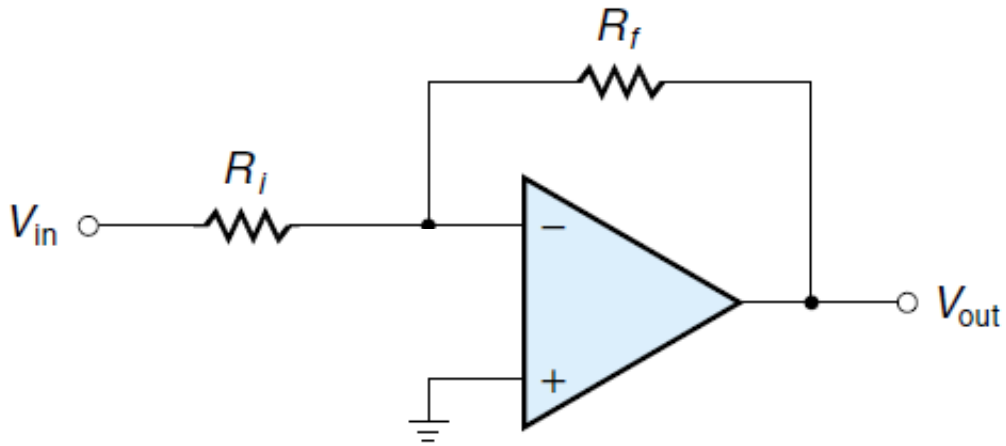
- Operational amplifier (Op-amp) can be used to **amplify the input signal** without affecting its other properties
- In order to use Op-Amp as an amplifier, we need to use a feedback type of circuit
- Resistors can be connected to Op-amp in a **negative feedback** manner to implement the desired gain



Voltage Follower

Inverting Amplifier

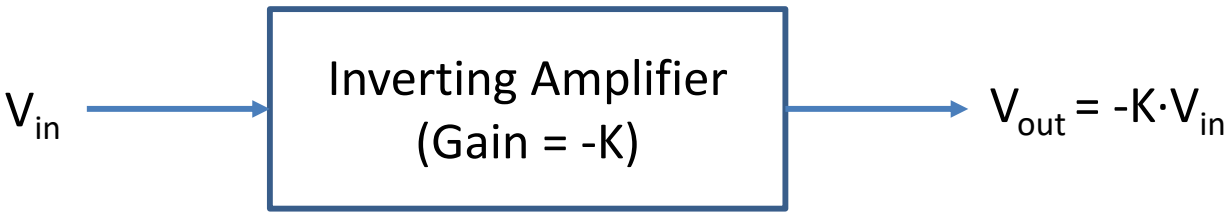
- Purpose: To multiply (and invert) a voltage signal by some factor



$$\text{Gain } A_{CL} = -\frac{R_f}{R_i}$$

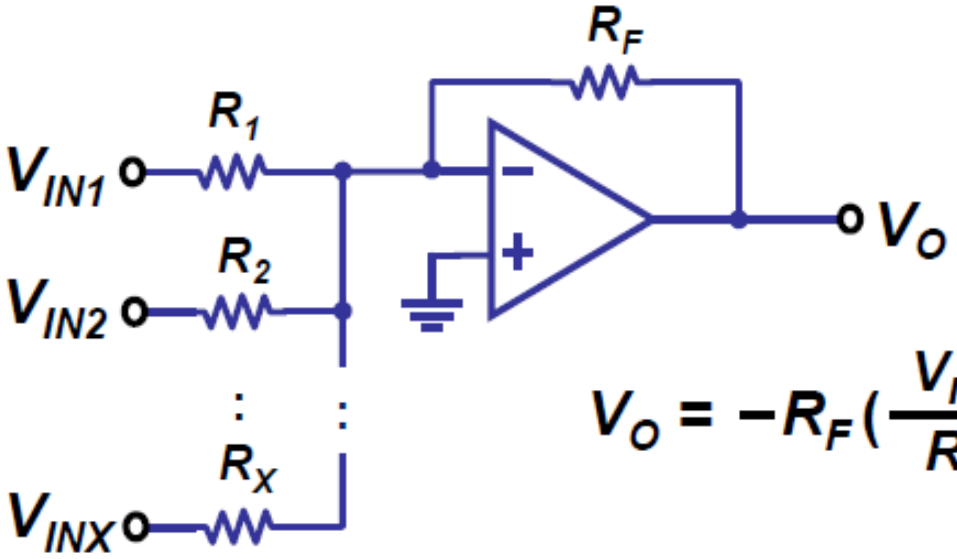
$$V_{out} = -\frac{R_f}{R_i} \times V_{in}$$

- Resistors R_f and R_i can be chosen to achieve any desired gain
  - If R_f is **greater** than R_i, the signal is **amplified** (and inverted)
  - If R_f is **less** than R_i, signal is **attenuated** (and inverted)



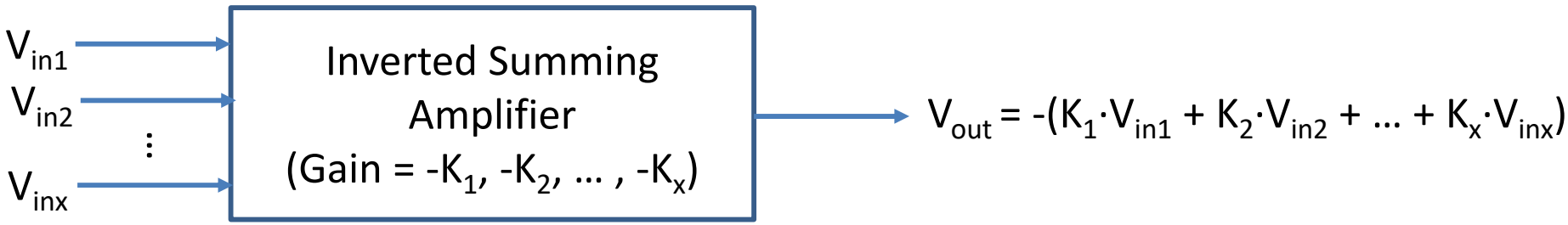
Inverted Summing Amplifier

- Purpose: To add multiple input signals together and produce a weighted sum output



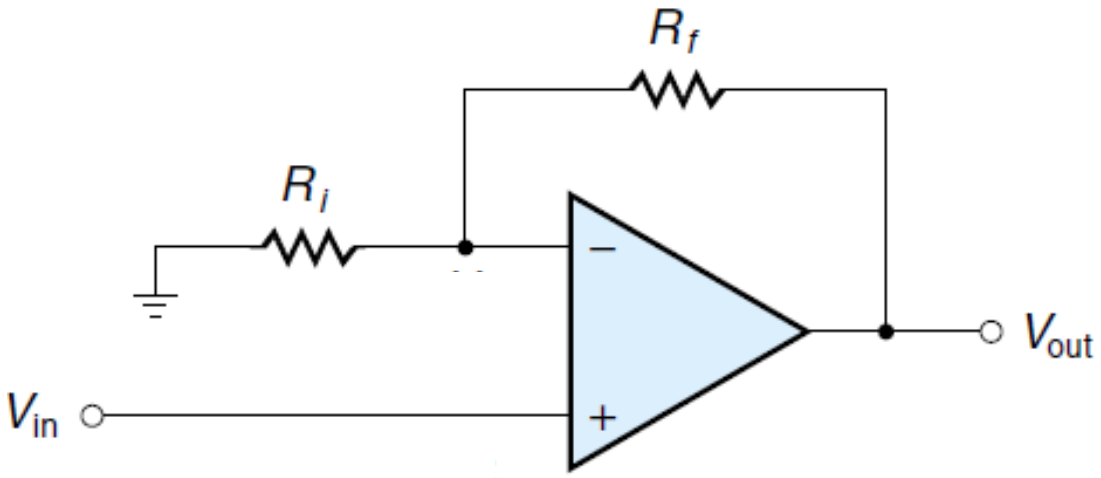
$$V_O = -R_F\left(\frac{V_{IN1}}{R_1} + \frac{V_{IN2}}{R_2} + \ldots + \frac{V_{INX}}{R_X}\right)$$

- The gain of each signal can be individually adjusted by choosing a different resistor value for each input
  - For example, to find the average of multiple signals



Non Inverting Amplifier

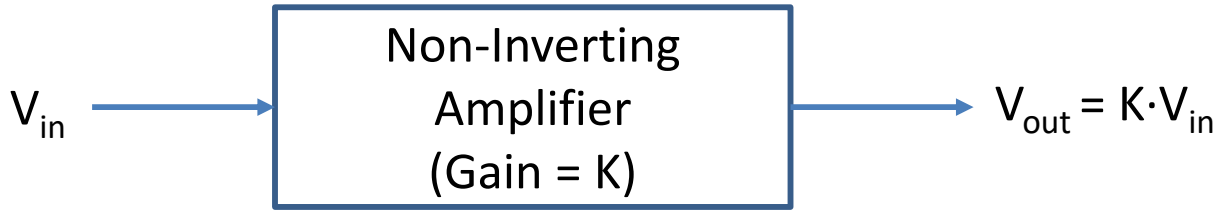
- Purpose: To amplify a voltage signal **without** inverting



$$\textbf{Gain } A_{CL} = \frac{R_f}{R_i} + 1$$

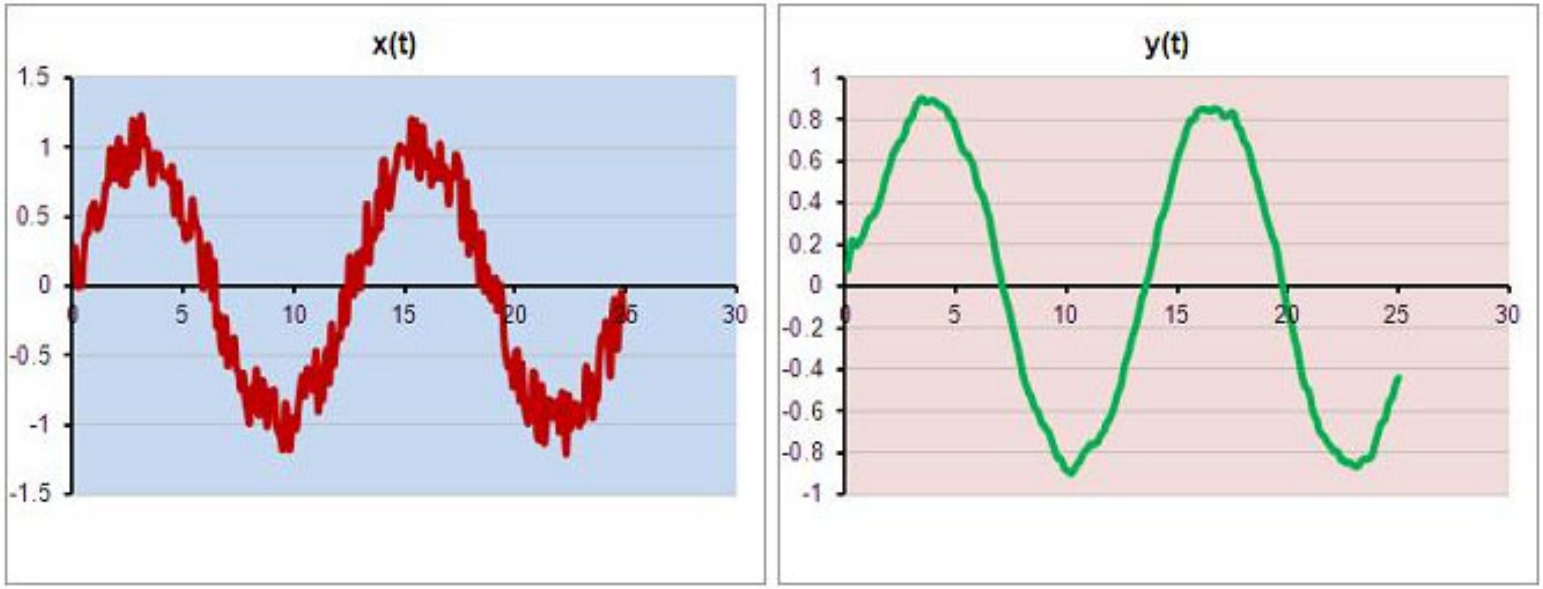
$$V_{out} = \left(\frac{R_f}{R_i} + 1\right) \times V_{in}$$

- Note that A_{CL} is always positive and always > 1



Why We Need Filters?

- Output signal of a sensor can become "contaminated" with unwanted signals or voltages that prevent the actual signal required from being measured correctly
- Noise is unwanted random addition to the signal and is typically at different (higher) frequency to the signal of interest
- These high frequency “Noise” can be either greatly reduced or even eliminated by using Low Pass Filters



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1 August 2022